

Data Exploration and Preparation

Diploma in DAB – L7

CCT College Dublin

Data Management and Exploratory Data Analysis

Week 3

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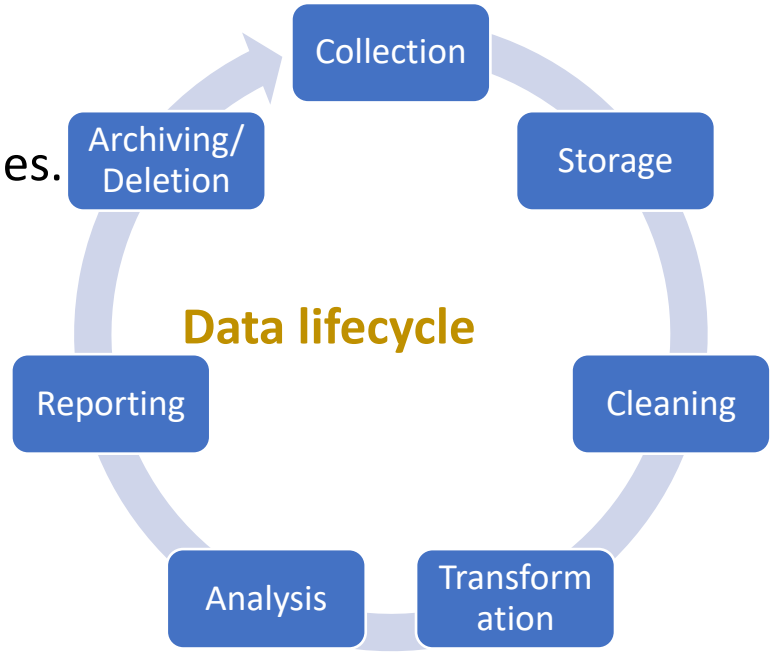
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Outline

- Data management
- Data consistency and documentation
- Data dictionaries, Data quality and preprocessing
- Legal, ethical and regulatory constraints
- Exploratory Data Analysis
- Data visualisation
- Statistical properties
- Univariate and multivariate analysis

Data Management Cycle

- **Data management** refers to the processes used to ***collect, store, organize, maintain, protect*** and ***use*** data effectively. Data management helps to ensure that data are
- **Accurate:** values correctly represent the real-world situation.
- **Complete:** All required information is present without any missing values.
- **Consistent:** the same definitions, formats and rules are used.
- **Accessible:** authorized users can locate and understand the data.
- **Secure:** data are protected from unauthorized access.
- **Documented:** meaning, origin and transformations are recorded.



- **Example:** If one system records gender as Male/Female, another uses M/F, and another uses 1/2, combining the datasets may produce inconsistencies unless these codes are documented and standardized.

Data Consistency

- **Data consistency** means that values, formats, definitions and relationships remain logically compatible throughout a dataset.
- **Requirement of Standard Formats**
  - Use validation rules
  - Establish naming conventions
  - Use controlled categories
  - Document changes
  - Check datasets before combining them
- **Main principle:** Consistent data make comparison, aggregation and analysis more reliable.

- **Common consistency problems**
- **Different date formats:**
  - 21/09/2026
  - 2026-09-21
  - Sep 21, 2026
- **Different units:**
  - kilograms versus pounds
  - euro versus dollars
- **Inconsistent category labels:**
  - Ireland
  - IRE
  - Republic of Ireland
- **Spelling differences:**
  - Dublin
  - DUBLIN
  - Dubln
- Duplicate records
- Different definitions for the same variable.

Documentation and the Data Dictionary

- A **data dictionary** is documentation that explains the structure, meaning and permitted values of variables within a dataset.

- **Characteristics**

- Interprets variables consistently
- Prevents incorrect assumptions
- Identifies data types and permitted ranges
- Makes datasets easier to share
- Supports reproducibility
- Helps future users to understand historical datasets.
- A good data dictionary should be updated whenever the dataset structure changes.

Typical Data-dictionary Fields

Variable	Description	Type	Example	Allowed values
Student_ID	Unique student identifier	Integer	10542	Positive integers
Age	Student age	Integer	24	18 - 100
Programme	Programme enrolled	Text	Computing	Defined programme list
Attendance	Attendance percentage	Decimal	86.5	0 - 100
Status	Student status	Category	Active	Active/ Inactive

Data Quality and Preprocessing

- Real-world datasets are not ready for analysis.
- Data preprocessing prepares raw data for reliable analysis. Common data-quality problems are mentioned below
  - Missing values
  - Duplicate observations
  - Incorrect data types
  - Invalid values
  - Inconsistent formatting
  - Typographical errors
  - Extreme or unusual observations
  - Inconsistent categories

Preprocessing Workflow

1. **Inspect the dataset:** Check rows, columns, variable types and dimensions.
2. **Identify problems:** Find missing, invalid or inconsistent values.
3. **Clean:** Correct errors, remove inappropriate duplicates and standardise formats.
4. **Transform:** Create useful variables or change representations.
5. **Validate:** Confirm that cleaning has not introduced additional problems.
6. **Document:** Record what was changed and why.

Data cleaning should not simply remove inconvenient observations; decisions should have a defensible reason.

Legal, Ethical and Regulatory Constraints

- Working with data involves responsibilities beyond the technical analysis.
- **Legal considerations**
- For organisations operating in Ireland and the EU, personal-data processing is governed principally by the **General Data Protection Regulation (GDPR)** and Ireland's **Data Protection Act 2018**. GDPR principles include
  - Lawfulness, fairness and transparency
  - Purpose limitation
  - Data minimisation
  - Accuracy
  - Storage limitation
  - Integrity and confidentiality
  - Accountability

- **Ethical considerations**
- Analysts should also consider
  - Privacy
  - Confidentiality
  - Bias and discrimination
  - Transparency
  - Representativeness
  - Potential harm
  - Responsible interpretation

Statistical Properties of Data

- Descriptive statistics summarise important characteristics of a dataset.
- **Measures of central tendency**
  - Mean, Median, Mode
- **Measures of spread**
  - Range, Variance, Standard deviation and Interquartile range (IQR)
- **Distribution properties**
- Analysts should also investigate the following properties
  - Shape, Symmetry , Skewness , Outliers , Minimum and maximum values and Quartiles
- **Example:** A large difference between the mean and median can indicate a skewed distribution or the influence of extreme observations.

Univariate Analysis

- **Univariate analysis** examines one variable at a time.
- **Numerical variable example:** Student Age
- Questions might include
- What is the average age?
- What is the median?
- What is the youngest student?
- What is the oldest student?
- How widely are ages distributed?
- Are unusual values present?
- **Useful techniques:** Mean , Median, Standard deviation, Minimum/maximum, Quartiles, Histogram and Box plot

- **Categorical example:** Course
- Questions might include:
- Which Course has the most students?
- How many students are in each category?
- What percentage belongs to each category?
- Useful techniques: Frequency tables, Percentages, Mode and Bar charts
- **What does this individual variable look like and discuss the properties?**

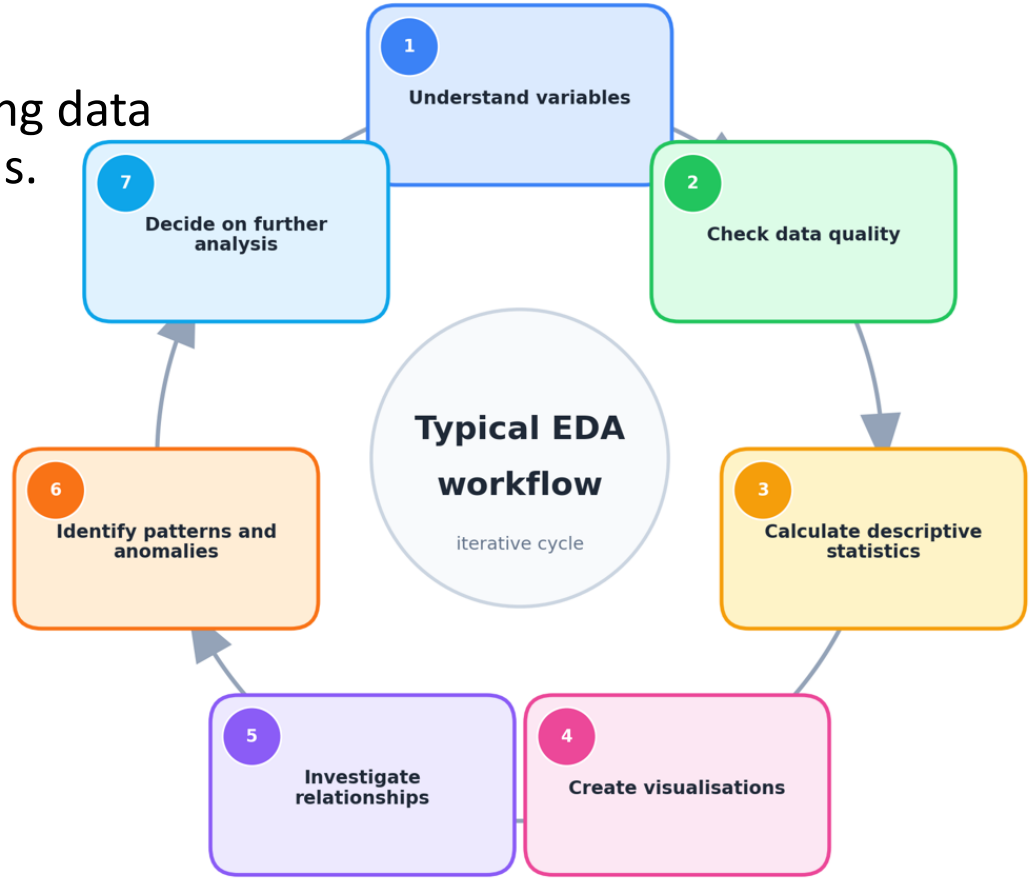
Bivariate and Multivariate Analysis

- **Bivariate analysis:** Examines the relationship between two variables. Examples are mentioned below
- **Numerical + Numerical**
- Study hours and examination results
- Scatter plot
- Correlation
- **Categorical + Numerical**
- Course and average examination result
- Grouped statistics
- Box plots
- **Categorical + Categorical**
- Programme and employment status
- Cross-tabulation
- Grouped bar chart

- **Multivariate analysis**
- Examines **three or more variables simultaneously**.
- Example:
- **Exam Result = Study Hours + Attendance + Previous Result + Programme**
- Useful approaches can include:
  - Correlation matrices
  - Scatterplot matrices
  - Grouped analysis
  - Multiple regression
  - Principal Component Analysis for more advanced datasets
- **Correlation does not automatically establish causation.**
- A relationship between variables may be influenced by additional variables or other factors.

Exploratory Data Analysis (EDA)

- **Exploratory Data Analysis (EDA)** is the process of examining data before applying formal models or drawing final conclusions.
- **Characteristics of EDA**
  - Understand the structure of the dataset
  - Identify missing information
  - Detect unusual observations
  - Explore distributions
  - Discover patterns
  - Investigate relationships
  - Test assumptions
  - Generate questions or hypotheses



EDA combines **numerical summaries and graphical exploration** rather than relying on only one method.

Visualisation in EDA

- Visualisation helps the data analysts to recognise patterns that may be difficult to identify from tables alone.
- **Bar chart:** Useful for categorical variables. For example, number of students by programme.
- **Histogram:** Shows the distribution of a numerical variable. For example, distribution of examination marks.
- **Box plot:** Shows median, quartiles, spread and potential outliers.
- **Scatter plot:** Examines relationships between two numerical variables. For example, study hours versus examination score.
- **Line chart:** Useful for analysing changes over time.

- **Visualisation Principles**
  - Choose a chart appropriate to the data type.
  - Give the chart a meaningful title.
  - Label axes clearly.
  - Include units.
  - Avoid unnecessary decoration.
  - Use consistent scales.
  - Avoid misleading axes or distorted representations.

Resources/ References

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